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Studierichting: Informatica
Oefeningenreeks 5I nr. 1.3

$$y = x^2 - \frac{1}{8}\ln x \Rightarrow y' = 2x - \frac{1}{8x} \Rightarrow (y')^2 = 4x^2 - \frac{1}{2} + \frac{1}{64x^2}$$

$$\Rightarrow 1 + (y')^2 = 1 + 4x^2 - \frac{1}{2} + \frac{1}{64x^2} = \frac{256x^4 + 32x^2 + 1}{64x^2}$$

$$\int_2^3 \sqrt{\frac{256x^4 + 32x^2 + 1}{64x^2}} dx = \int_2^3 \sqrt{\frac{\left(16x^2 + 1\right)^2}{64x^2}} dx = \int_2^3 \frac{16x^2 + 1}{8x} dx$$

$$= \int_2^3 \frac{16}{8}x + \frac{1}{8x} dx = \left[x^2 + \frac{1}{8}\ln|x|\right]_2^3 = 9 + \frac{1}{8}\ln|3| - \left(4 + \frac{1}{8}\ln|2|\right)$$

$$= 5 + \frac{1}{8}\ln\frac{3}{2}$$

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References